PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Factors associated with completion of maternity continuum of care in Ethiopia: A systematic review and meta-analysis
AUTHORS	mose, ayenew; Haile, Kassahun; Timerga, Abebe

VERSION 1 – REVIEW

REVIEWER	Jobi Babu Marian College Kuttikkanam, School of Social Work
REVIEW RETURNED	26-Mar-2022

GENERAL COMMENTS	The reviewer completed the checklist but made no further comments.

REVIEWER	Gizachew Tadele Tiruneh JSI Research & Training Institute, Inc., The Last Ten Kilometers (L10K) Project
REVIEW RETURNED	01-May-2022

GENERAL COMMENTS	It is my pleasure to be designed as a reviewer of this paper, thank you very much! This a good paper that examined the factors limiting the continuity of maternal health care in Ethiopia. 3. Data presentation: I would present determinant factors in Tabular form rather than multiple Funnel Plots Major comments 1. The definitions of COC vary across the studies included. Some articles defined COC as women who took ANC1, SBA, and PNC within 2 days or within 6 weeks; others defined as those who received ANC4+, SBA, and PNC within 2 days or within 6 weeks. Another paper also included continuation to at least 3 PNC visits. But this article took the definition of ANC4+, SBA, PNC with 6 weeks. This discrepancy in the definition of outcome variables would lead to variable prevalence of COC and determinant factors. As such, authors need to calculate the prevalence of COC by abstract appropriate numerators and denominators from the articles (either requesting the raw data from primary authors of the articles). 2. What is the theoretical background of this study? What variables are not determining COC and what variables not measured? What further studies is needed or what remains unanswered? I would like this paper to answer such questions too. Minor comments 1. On the abstract some determinants are directional others not. Make it consistent 2. Add any take home messages regarding determinants to the conclusion section of the abstract

	3. Introduction section, line 98, please describe inconsistent findings of the previous studies 4. use of PICO for cross-sectional non-interventional study designs without comparison group. Think of other themes like Condition, Context, Population, and Exposure 5. Analysis: Add narrative synthesis technique and 6. in Table 1 I would include other characteristics like study settings, the objective of the study, and if varied include recall period, enumerators educational background, mode and place of interview, etc. and narrate in the text and later in the Discussion section too. These would be source of variations... 7. Discussion section: Please summarize the main findings of the study in the first paragraph and then discuss each main findings in the subsequent paragraphs. Don't repeat the objective and start discussion before narrating the key findings 8. Line 248, EDHS surveys aren't reporting COC directly.

REVIEWER	Paola Rucci Alma Mater Studiorum – University of Bologna, Department of Biomedical and Neuromotor Sciences
REVIEW RETURNED	26-Jul-2022

GENERAL COMMENTS	The paper by Mose et al. is a meta-analysis of 9 studies carried out in Ethiopia on the prevalence of continuity of maternal care and its associated factors. Despite the key relevance of this public health topic, the paper confirms the established findings of the 9 meta-analysed studies in Ethiopia, and of a broader recently published study (Hunie Asratie M, Belay DG.Pooled Prevalence and Determinants of Completion of Maternity Continuum of Care in Sub-Saharan Africa: A Multi-Country Analysis of Recent Demographic and Health Surveys. Front Glob Womens Health. 2022 May 25;3:869552.) Thus, the lack of originality and generalization to other countries makes this manuscript more suitable for a specialized journal on women's health than for the BMJ open audience. In addition, this meta-analysis was not recorded in PROSPERO. Concerning specific points, the English form requires substantial editing. To mention a few: 'MIDLINE' instead of 'MEDLINE' Page 6 line 56 'the secondary outcome was' should be 'the secondary aim was to identify' Page 6 line 46 . 'Therefore, all included studies were asked the study participants' change to 'in all included studies the study participants were asked….' Page 8, line 50 "As a result, 9 studies were fulfilled the inclusion criteria to undergo the final meta-analysis" Ref 25 should be completed with bibliographic information Page 3, line 39 The following statement is questionable: "All included studies were cross sectional which might be unable to infer a causal relationship between variables", because the research question requires a cross-sectional design. Page 6, line 48. The meaning of the acronyms 4+ ANC, SBA in the health facility, and PNC is not provided.

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer #1

Prof. Jobi Babu, Marian College Kuttikkanam

Comments to the Author: nil

Author response: Dear Prof. Jobi Babu, Thank you for your time and consideration.

Response to Reviewer #2

Dr. Gizachew Tadele Tiruneh, JSI Research & Training Institute, Inc.

Comments to the Author:
It is my pleasure to be designed as a reviewer of this paper, thank you very much! This a good paper that examined the factors limiting the continuity of maternal health care in Ethiopia.
3. Data presentation: I would present determinant factors in Tabular form rather than multiple Funnel Plots

Author response: Dear Dr. Gizachew Tadele, thank you very much for giving your time to review this manuscript. Saying this, I have putted all funnel plots in a single table forum (Table 3).

Major comments

1. The definitions of COC vary across the studies included. Some articles defined COC as women who took ANC1, SBA, and PNC within 2 days or within 6 weeks; others defined as those who received ANC4+, SBA, and PNC within 2 days or within 6 weeks. Another paper also included continuation to at least 3 PNC visits. But this article took the definition of ANC4+, SBA, PNC with 6 weeks. This discrepancy in the definition of outcome variables would lead to variable prevalence of COC and determinant factors. As such, authors need to calculate the prevalence of COC by abstract appropriate numerators and denominators from the articles (either requesting the raw data from primary authors of the articles).

   Author response: Thank you for raising these critical questions. I have reviewed all included studies in detail. I have made mistake during operationalizing the outcome variables. Therefore, I have corrected the definitions of the outcome variables as each pregnant woman has one or more antenatal care (ANC) contacts, birth attended by skilled health personnel, and receive early routine postnatal care within 2 days up to six weeks). Moreover, I have discussed it in the limitation of the study section.

2. What is the theoretical background of this study? What variables are not determining COC and what variables not measured? What further studies are needed or what remains unanswered? I would like this paper to answer such questions too.

Author response: Thank you in Advance. I have briefly described the theoretical background of this study in the introduction section. Let me explain here 1/ improving completion of the maternity continuum of care (4 and above (ANC) contacts, birth attended by skilled health personnel, and receives early routine postnatal care within 2 days) is crucial for the health of women, neonates, and child to achieve the sustainable development goal by 2030(i.e. end preventable cause of maternal and neonatal mortality). However, in low resource setting including Ethiopia several evidences shows that there is poor utilization of this service. Therefore, this study dig out factors associated with completion of maternity continuum of care which is helpful for policy makers to design strategies that

can enhance completion of maternity continuum of care in the respective region. 2/ for the second questions, what variables are not determining COC and what variables not measured?, According to the result of meta-analysis the factors that not determine COC were rural residence, lack of educational status, not utilizing pre-pregnancy contraceptive utilization, lack of women's decision, not following mass media, not early initiation of ANC, did not have planned pregnancy, and long distance from health facility. For the third questions What further studies are needed or what remains unanswered? Based on data bases searches we did not find studies conducted in Afar, Gambella, Somalia, and Tigray region. Therefore, we would like to recommend primary studies in the above listed regions.

Minor comments

1. On the abstract some determinants are directional others not. Make it consistent

   Author response: Thank you in advance. I have corrected it. For instance, 'Autonomy' changed into 'Women Autonomy' and 'distance from a health facility' changed into 'distance from a health facility <30' etc.
2. Add any take home messages regarding determinants to the conclusion section of the abstract

   Author response: Thank you. Here is the take home message I have added in the revised manuscript. ''Improving accessibility of health facility, women's awareness, and empowering women's decision making are recommended to enhance completion of maternity continuum of care.''
3. Introduction section, line 98, please describes inconsistent findings of the previous studies.

   Author response: Thank you very much. Regarding inconsistence findings, for instance, a study conducted by Shitie A. et al in Amhara region reported that 42% of women had complete maternity continuum of care [15]. However, a study conducted by Haile D, et al in southern region reported that only 9.7% of women had complete maternity continuum of care [17].
4. Use of PICO for cross-sectional non-interventional study designs without comparison group. Think of other themes like Condition, Context, Population, and Exposure

   Author response: Thank you in advance. In the information source, I have also used the exposure that may explain or predict the presence of a study outcome. The outcome variable (completion of maternity continuum of care).
5. Analysis: Add narrative synthesis technique and

   Author response: Thank you in Advance. I have used tables, specifically for database searches based on the editor comment.
6. In Table 1 I would include other characteristics like study settings, the objective of the study, and if varied include recall period, enumerators educational background, mode and place of interview, etc. and narrate in the text and later in the Discussion section too. These would be source of variations...

Author response: Thank you very much. I have added these important characteristics including study setting, objective of the study, sampling procedure, and mode of interview in the table, text and discussion section.

7. Discussion section: Please summarize the main findings of the study in the first paragraph and then discuss each main finding in the subsequent paragraphs. Don't repeat the objective and start discussion before narrating the key findings

   Author response: Thank you in advance. I have done it in the revised manuscript. I have putted the key findings in the first paragraph of discussion section ''the current study revealed that the estimated prevalence of completion of maternity continuum of care was 25.82%. Urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness and complication readiness, distance from health facility less than 30 minutes were factors associated with completion of maternity continuum of care in Ethiopia.''

8. Line 248, EDHS surveys aren't reporting COC directly.

   Author response: Thank you in advance. I have removed the statement that related to EDHS.

Response to Reviewer #3

Dr. Paola Rucci, Alma Mater Studiorum – University of Bologna

Comments to the Author:

The paper by Mose et al. is a meta-analysis of 9 studies carried out in Ethiopia on the prevalence of continuity of maternal care and its associated factors.

Despite the key relevance of this public health topic, the paper confirms the established findings of the 9 meta-analysed studies in Ethiopia, and of a broader recently published study (Hunie Asratie M, Belay DG. Pooled Prevalence and Determinants of Completion of Maternity Continuum of Care in Sub-Saharan Africa: A Multi-Country Analysis of Recent Demographic and Health Surveys. Front Glob Womens Health. 2022 May 25;3:869552.)

Thus, the lack of originality and generalization to other countries makes this manuscript more suitable for a specialized journal on women's health than for the BMJ open audience.

In addition, this meta-analysis was not recorded in PROSPERO.

Author response: Dear Dr. Paola Rucci, I would like to thank you very much for giving your time to review this manuscript. I have accepted your comments and questions. Thus, below I have putted my response point by point. Thank you in in advance. Here is the PROSPERO **record Id: CRD42022312692**. Regarding the originality of this study, you cited one article which was used secondary data (EDHS) and they were considered at least one ANC visit for the completion of the maternity continuum of care. However, our review considered 4 and above antenatal care visits which is significant difference. Moreover, I think it is suitable for BMJ Open audience because, one of the

main objectives of BMJ open journal is dedicated exclusively to publishing medical research including the vulnerable group of populations. Thus, I will say that articles in the field of women health can be published in this journal.

Concerning specific points, the English form requires substantial editing. To mention a few:

'MIDLINE' instead of 'MEDLINE'

Author response: Thank you in advance. I have corrected it in the revised version of the manuscript/ used PubMed in the entire manuscript. Moreover, I have gone through each paragraph of the manuscript and corrected all typos.

Page 6 line 56 'the secondary outcome was' should be 'the secondary aim was to identify'

Author response: Thank you in advance. I have corrected it in the revised version of the manuscript.

Page 6 line 46. 'Therefore, all included studies were asked the study participants' change to 'in all included studies the study participants were asked….'

Author response: Thank you in advance. I have corrected it in the revised manuscript.

Page 8, line 50 "As a result, 9 studies were fulfilled the inclusion criteria to undergo the final meta-analysis"

Author response: Thank you in advance. I have corrected it

Ref 25 should be completed with bibliographic information

Author response: Thank you in advance. I have corrected it. 'Tsega D, Admas M, Talie A, Tsega TB, Birhanu MY, Alemu S, Mengist B. Maternity Continuum Care Completion and Its Associated Factors in Northwest Ethiopia. Journal of Pregnancy. 2022 Feb 17;2022''**.**

Page 3, line 39 The following statement is questionable: "All included studies were cross sectional which might be unable to infer a causal relationship between variables", because the research question requires a cross-sectional design.

Author response: Thank you in advance. I have removed this statement and replace by other limitations that I have found in the methodology section.

Page 6, line 48. The meaning of the acronyms 4+ ANC, SBA in the health facility, and PNC is not provided.

Author response: Thank you in advance. I have briefly described the meaning of 4+ ANC, SBA, and PNC. Below, I have written the statement that I made correction 'Firstly, having greater than or equal to 4 antenatal care visits. Antenatal care is a type of preventive healthcare which is provided in a

regular check-up that allow doctors or midwives to treat and prevent potential health problems throughout the course of the pregnancy. Secondly, give birth at health facilities with skilled birth attendant (SBA). Skilled birth attendant is a midwife, physician, and obstetrician who provide essential and emergency health care services to women and their newborns during pregnancy, and childbirth. Thirdly, receive postnatal care after delivery. It is defined as specialized care for the mother that begins immediately after delivery, up to a period of six weeks. The care includes prevention, elimination, early detection and cure of health complications, counselling on breastfeeding, and immunization.''

VERSION 2 – REVIEW

REVIEWER	Gizachew Tadele Tiruneh JSI Research & Training Institute, Inc., The Last Ten Kilometers (L10K) Project
REVIEW RETURNED	07-Sep-2022

GENERAL COMMENTS	It has significantly improved from the first version. Minor comments include; Outcome variable Rephrase the “Outcome measurements” section. Studies could not directly interview respondents on the completion COC as described in lines 150-151 and 160-162 rather researchers would measure from the subsequent use of ANC4, SBA, and PNC services. You might define the outcome variable as “The outcome variables of interest of the study were the uptake of the CoC at antepartum, intrapartum, and postpartum stages: 1) continuum of care at the antepartum stage is women who received four or more antenatal care (ANC4+) visits, 2) continuum of care at the intrapartum stage is those women who continued use of skilled birth attendance after receiving ANC4+ visits, and 3) continuum of care at the postpartum stage or complete continuum is those women who received PNC for the mothers and their newborns, within six weeks of their delivery after receiving both ANC4+ visits and delivered by skilled assistance.” And then indicate that researchers included studies that fulfill this definition or do a subgroup analysis based on the definition of COC to show whether the discrepancy in the definition of outcome variable would lead to variables in prevalence of COC and/or determinant factors or not. Analysis Please add narrative analysis of the qualitative data and a random effect model for quantitative analysis. You might add “A narrative synthesis was used to analyze and interpret the findings. A random-effects meta-analysis model was used to pool the estimates of the magnitude of COC and its determinant factors accounting for the variability among studies.” Subgroup analysis: In addition to mentioning differences between studies in defining COC as limitations of the study, authors could do subgroup analysis or drop studies that could not fulfil the definition Labeling of figures and tables

	□ Line 210 change “Table 1.” to “Table 1:”; line 231 change “Table, 2 sensitivity…” to “Table 2: Sensitivity…” □ Line 213, change “C” to “CS” □ Change “Fig 1;” to “Figure 1:”; “Fig 2;” to “Figure 2:”; etc. □ Change “Supplementary File 1;” to “Supplementary File 1:”; “Supplementary File 3;” to “Supplementary File 3:” □ and “Meta-analysis” as keywords Grammar and editorial □ Line 109, drop the phrase “In conclusion” or change from concluding phrase to reason marker like therefore, accordingly, consequently, etc. □ Line 169, drop the phrase “the using the” □ Line 269 the sentence “The pooled prevalence of completion of maternity continuum of care was 25.82%.” is repeated in the first paragraph. Paraphrase it or interprete it like “the prevalence of completion of maternity continuum of care was low in Ethiopia” □ Line 529, the sentence is incomplete Keywords □ often did not contain study designs or analytic techniques. I would drop keywords “Systematic review” and “Meta-analysis”

VERSION 2 – AUTHOR RESPONSE

Response to Reviewer #2

Dr. Gizachew Tadele Tiruneh, JSI Research & Training Institute, Inc.

Comments to the Author:

It has significantly improved from the first version. Minor comments include;

Author response: Dear Dr. Gizachew Tadele, thank you very much for reviving this manuscript. And I found that your comments were acceptable. Therefore, I have corrected all of your concerns in the revised manuscript. Thank you once again for your time and consideration.

Outcome variable

1. Rephrase the “Outcome measurements” section. Studies could not directly interview respondents on the completion COC as described in lines 150-151 and 160-162 rather researchers would measure from the subsequent use of ANC4, SBA, and PNC services. You might define the outcome variable as “The outcome variables of interest of the study were the uptake of the CoC at antepartum, intrapartum, and postpartum stages: 1) continuum of care at the antepartum stage is women who received four or more antenatal care (ANC4+) visits, 2) continuum of care at the intrapartum stage is those women who continued use of skilled birth attendance after receiving ANC4+ visits, and 3) continuum of care at the postpartum stage or complete continuum is those women who received PNC for the mothers and their newborns, within six weeks of their delivery after receiving both ANC4+ visits and delivered by skilled assistance.”

Author response: Thank you for your suggestion. I have rewritten this section as ‘The outcome variables of interest of the study were the uptake of the CoC at antepartum, intrapartum, and postpartum stages: 1) continuum of care at the antepartum stage is women who received four or more antenatal care (ANC4+) visits, 2) continuum of care at the intrapartum stage is those women who continued use of skilled birth attendance after receiving ANC4+ visits, and 3) continuum of care at the postpartum stage or complete continuum is those women who received PNC for the mothers

and their newborns, within six weeks of their delivery after receiving both ANC4+ visits and delivered by skilled assistance [7,8]. The secondary aim was to identify factors associated with the completion of the maternity continuum of care.

2. And then indicate that researchers included studies that fulfil this definition or do a subgroup analysis based on the definition of COC to show whether the discrepancy in the definition of outcome variable would lead to variables in prevalence of COC and/or determinant factors or not.

Author response: Thank you in advance; we have included 9 studies which fulfilled the operational definition. Moreover, we have performed subgroup analysis based the study region. However, we were not performed Subgroup analysis based on the definition of COC. Since, we were not identified significant difference on the definition of COC.

Analysis

3. Please add narrative analysis of the qualitative data and a random effect model for quantitative analysis. You might add "A narrative synthesis was used to analyze and interpret the findings. A random-effects meta-analysis model was used to pool the estimates of the magnitude of COC and its determinant factors accounting for the variability among studies."

Author response: Thank you in advance. In this review, we were not included qualitative studies. However, I have added the statement *"A narrative synthesis was used to analyze and interpret the findings. A random-effects meta-analysis model was used to pool the estimates of the magnitude of COC and its determinant factors accounting for the variability among studies"* in the data processing and analysis section.

4. Subgroup analysis: In addition to mentioning differences between studies in defining COC as limitations of the study, authors could do subgroup analysis or drop studies that could not fulfil the definition.
   Author response: thank you in advance we have done subgroup analysis. And we were not included studies which did not fulfil the inclusion criteria and outcome measurements definition.

Labelling of figures and tables

5. Line 210 change "Table 1." to "Table 1:"; line 231 change "Table, 2 sensitivity…" to "Table 2: Sensitivity…"

Author response: Thank you very much. Yes, I have changed it in the revised manuscript.

6. Line 213, change "C" to "CS"

Author response: Thank you in Advance, I have changed it.

7. Change "Fig 1;" to "Figure 1:"; "Fig 2;" to "Figure 2:"etc.

Author response: Thank you in advance. I have corrected it.

8. Change "Supplementary File 1;" to "Supplementary File 1:"; "Supplementary File 3;" to "Supplementary File 3:"and "Meta-analysis" as keywords

Author response: Thank you very much. Yes I have changed it in the revised manuscript.

Grammar and editorial

9. Line 109, drop the phrase “In conclusion” or change from concluding phrase to reason marker like therefore, accordingly, consequently, etc.

Author response: Thank you. I have used the word ‘accordingly’.

10. Line 169, drop the phrase “the using the”

Author response: **Thank you. I have dropped it in the revised manuscript.**

11. Line 269 the sentence “The pooled prevalence of completion of maternity continuum of care was 25.82%.” is repeated in the first paragraph. Paraphrase it or interprete it like “the prevalence of completion of maternity continuum of care was low in Ethiopia”

Author response: Thank you in advance for your suggestion. I have used this statement ‘the prevalence of completion of maternity continuum of care was low in Ethiopia.

12. Line 529, the sentence is incomplete

Author response: Thank you in Advance. It was typo problems, I have removed it.

Keywords

13. Often did not contain study designs or analytic techniques. I would drop keywords “Systematic review” and “Meta-analysis”

Author response: Thank you in advance. I have dropped those two key words in the revised manuscript and replaced with *Antenatal care, skilled birth attendants, and postnatal follow-up.*

VERSION 3 – REVIEW

REVIEWER	Gizachew Tadele Tiruneh JSI Research & Training Institute, Inc., The Last Ten Kilometers (L10K) Project
REVIEW RETURNED	30-Oct-2022

GENERAL COMMENTS	All the comments are adequately addressed. You might change title into "Prevalence of completion of maternity continuum of care and its associated factors in Ethiopia: A systematic review and meta-analysis" as the primary outcome of the study was determining pooled prevalence 2. If Tables are published as it is, you might format by remove vertical lines, this is often the standard Table and Figure formatting of journal articles